THE COZY SCIENCE LAB · ORIGINAL STUDY SERIES

Science Skills Lab

A Grade 7–10 study handbook with concept notes, examples, self-check explanations and labelled illustrations.

In this handbook

- Scientific Method & Variables
- Measurement & SI Units
- Data, Graphs & Lab Safety

Edition 1 · Lab-authored learning materials. Supporting publications are credited with each topic.
Examples are constructed for teaching; diagrams state their model and scale limits. For current Philippine hazard or weather decisions, consult the responsible agencies rather than a static study guide.

Scientific Method & Variables

Scientific investigation connects questions, evidence and explanations. It is iterative, not a single rigid checklist for every kind of science.

Ask something testable

A useful question states what will be observed or compared. Define how a variable will be measured so another investigator can understand or repeat the work.

Design a fair comparison

An independent variable is deliberately changed in a simple experiment; a dependent variable is measured. Controlled conditions help reduce alternative explanations. Observational studies also provide scientific evidence when experiments are not suitable.

Evaluate evidence

Repeated measurements, comparison groups when appropriate, uncertainty and possible bias affect interpretation. A result can support an explanation without proving it beyond all possible revision.

Key terms

Hypothesis

A testable proposed explanation or prediction within an investigation.

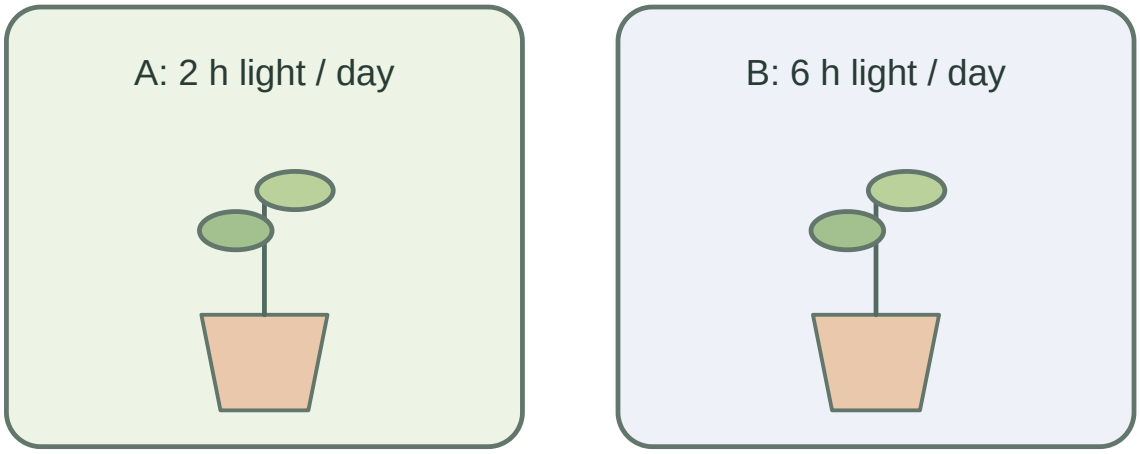
Controlled variable

A condition held consistent to make a comparison more informative.

Light and plant growth

Compare light exposure while using comparable plants and keeping water, soil, pot size and duration consistent. Measure growth with a defined method. State the limits of the comparison.

Remember: Change one variable at a time is useful for simple controlled experiments; it is not a claim that all valid science must use that exact design.



Measure growth using the same method

Same species, soil, pots, water and observation time

Design a fair comparison. Constructed teaching example, not experimental results. Use replication and consider confounding variables in a real investigation.

Check your thinking

Why report how you measured growth, not just 'the plants grew more'?

Explanation: A defined measurement makes the evidence interpretable and helps another person repeat or evaluate the comparison.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

OpenStax · Rice University. Biology 2e. https://openstax.org/details/books/biology-2e

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf

Measurement & SI Units

A measurement combines a value, a unit and an understanding of uncertainty. Good records make comparisons possible.

SI foundations

The seven SI base units are metre (m), kilogram (kg), second (s), ampere (A), kelvin (K), mole (mol) and candela (cd). Derived units combine base units; for example, speed can be measured in m/s.

Converting units

A prefix changes scale: centi means 10^{-2} and milli means 10^{-3}. Multiply by an equivalent unit ratio so unwanted units cancel. Keep units visible rather than treating them as decoration.

Precision and accuracy

Precision concerns agreement or spread among repeated measurements. Accuracy concerns closeness to a reference value. Instrument resolution, calibration and procedure all affect what a measurement can support.

Key terms

Resolution

The smallest displayed or distinguishable change on an instrument.

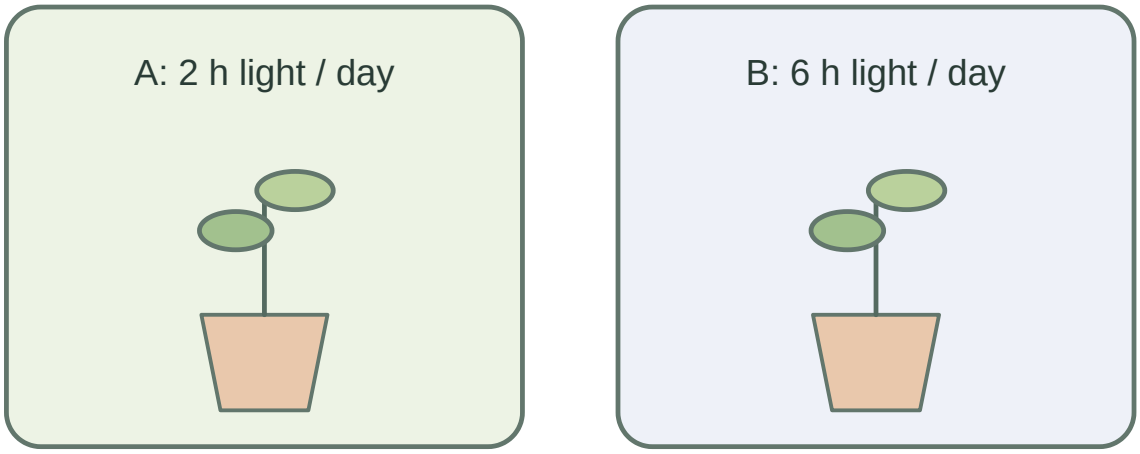
Uncertainty

A quantified or described limitation in knowledge of a measurement value.

Centimetres to metres

2.5 cm × (1 m / 100 cm) = 0.025 m. The centimetre units cancel, and the smaller numerical value is consistent with using a larger unit.

Remember: Mass and weight use different units and describe different quantities. More digits in a display do not automatically mean greater accuracy.



Design a fair comparison. Constructed teaching example, not experimental results. Use replication and consider confounding variables in a real investigation.

Check your thinking

Can repeated results be precise but inaccurate?

Explanation: Yes. They may agree closely with one another while a calibration error shifts all of them away from a reference value.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

Bureau International des Poids et Mesures. The International System of Units (SI). https://www.bipm.org/en/publications/si-brochure

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf

Data, Graphs & Lab Safety

Organized data help make evidence understandable. A clear graph and a safe method are both parts of useful science.

Choose a display

Bar charts compare categories, line graphs can show ordered changes, and scatter plots help examine associations. Label axes and units, state what was measured and choose scales that do not mislead.

Describe a pattern carefully

A mean is the sum divided by the number of values; a median is the middle of sorted data. Outliers can affect summaries. An association on a graph does not by itself show that one variable causes another.

Work safely

Follow the teacher's procedure and use appropriate protective equipment. Read labels, keep food away from lab work, never taste unknown substances or pipette by mouth, and report spills or damaged equipment promptly.

Key terms

Mean

The arithmetic average of a set of numerical values.

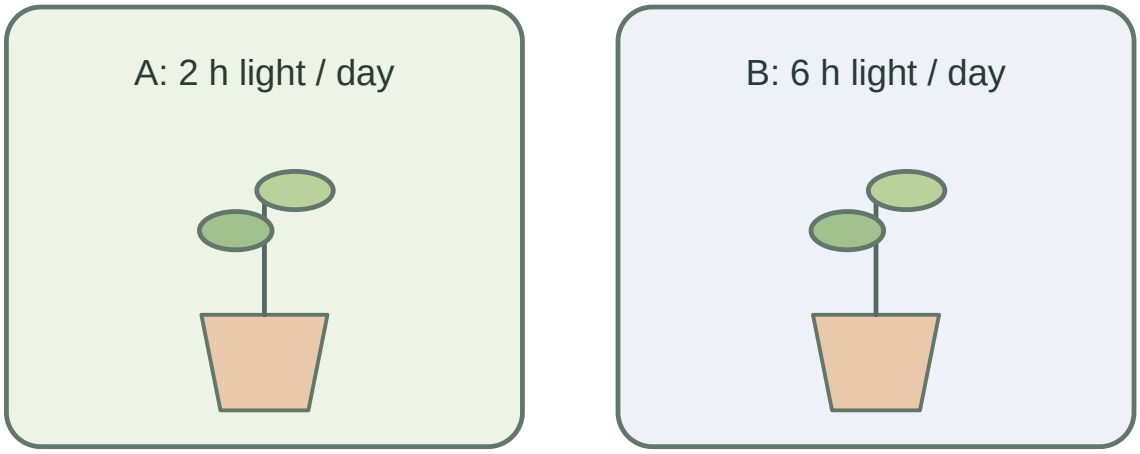
Correlation

An association between variables that does not alone establish causation.

A small dataset

For 2, 3 and 10, the mean is 5 and the median is 3. Both calculations are correct, but they emphasize different aspects of the data.

Remember: A steep-looking line can result from the chosen axis scale. Inspect labels and numbers before interpreting the visual impression.



Measure growth using the same method

Same species, soil, pots, water and observation time

Design a fair comparison. Constructed teaching example, not experimental results. Use replication and consider confounding variables in a real investigation.

Check your thinking

Why is 'these two quantities rise together' insufficient to establish a cause?

Explanation: Other variables, chance or reverse direction can help explain the association. A stronger design and additional evidence are needed.

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OpenStax · Rice University. Introductory Statistics 2e. https://openstax.org/details/books/introductory-statistics-2e

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf